

Mobile Application Development Technology (3-0-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Objectives:

1. To provide students the knowledge of recent trends in mobile application development
2. To give the overview of existing mobile operating system and the development SDKs required to develop application based on them
3. To reach students the basic of application development with reference to android environment

Course Contents:

1. Introduction to mobile Oses

7 hrs

- 1.1 Introduction of Mobile Oses: Android, iOS, Ubuntu, Touch, Blackberry, Tizen, Firefox OS, Symbian, Windows Phone
- 1.2 Build and Structures of Mobile Oses
- 1.3 Introduction to development environment (Native v/s HTML5)
- 1.4 Introduction to Android
 - 1.4.1 API levels /versions of Android
 - 1.4.2 Pros and Cons of Android
 - 1.4.3 Comparison of Android with other Mobile Oses
- 1.5 Introduction to Android VM and Runtime (Dalvik and ART)
- 1.6 Installation and configuration of Android SDKs and Eclipse IDE
 - 1.6.1 Their integration using ADT Plug-in
 - 1.6.2 Running an Emulator
- 1.7 Using ADB Command Line Interface

2. Android Classes and Basic's

9 hrs

- 2.1 Android Fundamentals
 - 2.1.1 Creating an Android App
 - 2.1.2 Android Manifest File
- 2.2 The Activity Class
 - 2.2.1 Activity Lifecycle
 - 2.2.2 Extending the Activity Class
 - 2.2.3 Creating Default Activity
 - 2.2.4 Creating Splash and Login Activities
- 2.3 The Intent class
 - 2.3.1 Creating Intent
 - 2.3.2 Switching between Activities using Intent
- 2.4 Permissions
 - 2.4.1 Allow App Permission in Android Manifest
- 2.5 The Fragment Class and Its uses

3. Android User Interface	8 hrs
3.1 Introduction to Multiple Screen Size and Orientation Interfaces	
3.2 User Interface Classes	
3.3 Android XML Layouts, Resource and Styles	
3.4 Android 3 rd Party UI/UX Libraries	
4. Advance Topics	6 hrs
4.1 User Notification	
4.2 The Broadcast Receiver class	
4.3 Threads, AsyncTask and Handlers	
4.4 Alarms	
4.5 Networking	
5. Graphics and Multimedia	6hrs
5.1 Graphics and Animations	
5.2 Multi-touch and Gestures	
5.3 Multimedia	
6. API's and Sensors	4hrs
6.1 Sensors	
6.2 Camera API	
6.3 Location and Map API	
6.4 Device Manager API	
6.5 Play Service API	
6.6 Multi-Player Gaming API	
7. Packaging and Monetizing	8 hrs
7.1 Data Management (Using SQLite database, local storage)	
7.2 The Content provider classes	
7.3 The Service classes	
7.4 Google Mobile Ads SDK	
7.5 Signing and Exploring an App	
7.6 Publishing your App to the Play Store	

Laboratory:

The laboratory classes should be based on the topics covered in the lecture classes that should give the students a hands-on training and familiarize them with the development environment

The laboratory works should be enable students to setup and run their own development environment and provide them with the base for their application development and publishing. It should also focus on publishing the application to the App Store and also monetize the apps using one of the advertising network provider y the SKD

Text Book:

1. Zigurf Mednieks: Liard Dornin, G. Blake Meike, Mausami Nakamura “Programming Android: Java Programming for the New Generation of Mobile Devices”, 2nd Edition, O’Reilly 2012
2. Reto Meier, “Professional Android Application Development”, Weily Publishing Inc, 2009, ISBN: 978-0-470-344710

Reference Books:

1. Barry Burd, Android Application Development All-inOne for Dummies, John Wiley & Sons, Inc 2012